



Analysis of Burnup effects and Its Integrity Assessment in the Interim of Irradiation with Molecular Dynamics

Ahli K.D. Willie^a, Hongtao Zhao^{a,b*}, M. Mustafa Azeem^a, Teplinskaya Svetlana^c

^a Fundamental Science on Nuclear Safety and Simulation Technology Laboratory, Harbin Engineering University, Harbin, 150001, China

^b Institute of Technical Physics, Heilongjiang Academy of Sciences 150086

^c College of Material Science and Chemical Engineering, Harbin Engineering University, Harbin, 150001, China

Abstract:

Burnups can cause major structural changes in the edge of the fuel rod and a general degradation of the thermal conductivity. In irradiated mixed oxide fuels of UO_2 , PuO_2 with NpO_2 as fission products (FP) various chemical states depending on the conditions of the fuel is developed. This work, we firstly applied the MD relation to obtain the thermal conductivity of UO_2 , PuO_2 , and (U, Pu) O_2 in temperature range of 300–2000 K. Lattice parameter, Burnup and the thermal conductivity were then calculated for specified UO_2 and PuO_2 . This calculation relates the degradation of thermal conductivity with a number of pores and increasing temperature. Finally, the migration energy barrier and the recovery energies of the obstruction type defects were calculated with molecular dynamics simulation.

Key words: burnup, Molecular dynamics, thermal conductivity

e-mail: willie@hrbeu.edu.cn

INTRODUCTION

During the operation of a nuclear reactor, the structural materials are exposed to high levels of radiation doses and high temperatures during their operational lifetime in reactors [1-7]. Uranium-base fuel results in changes in the composition of the fuel system due to loss of fissionable materials as a result of fission and the buildup of fission products.

Subjecting the mixed oxide (MOX) fuel in-reactor environment leads to microstructural changes due to the evolution of radiation produced defects, segregation and precipitation of fission products, and mechanical deformation. Consequently, little minor actinide (Np, Am, Cm) elements are generated due to the process of neutron capture and the transmutation products which will influence the thermal properties of mixed oxide fuels in turn.

In view of reactor fuel property evaluation, over the past decade there has been increased international recognition of the need for expanded, high quality, public sources of burnup data to validate the U-MOX spent fuel calculations [8]. Measurement of thermal properties under high temperature conditions is difficult to perform because of the cost of expensive appliances and materials as well as the extensive safety precautions, though it is important to evaluate comprehensively the thermal properties of mixed oxide fuel. In view of this, classical molecular dynamics (MD) is an efficient simulation technique to obtain the information regarding thermal properties of MOX over a wide temperature range.

Activation energies calculation and the importance of empirical pair potentials can vary strongly depending on the choice of potential. Govers et al. examined three different potentials for UO_2 (those of Basak [9], Jackson [10] and Morelon [11]) coupled with different parameterizations for the U-Xe and O-Xe interactions from Geng [12] and Nicoll [13] and recommend values of 6.5 eV, 4.5 eV and 2.4 eV [14] for the different stoichiometric regimes in very good agreement with the experimental values of 6.0 eV, 3.9 eV and 1.7 eV respectively [15]. As burnup is increased there is a marked increase in the dislocation density [16] in the fuel, that is, until the formation of the high burnup structure [17]. Clearly, the accuracy of MD simulation is directly determined by the interatomic potential used. Therefore understanding and selection is key to designing around their detrimental effects and to developing new alloys that are more radiation tolerant.

Physical properties of UO_2 [18, 19, 20] [21], PuO_2 [22][23], UO_{2-x} [24], PuO_{2-x} [25], $(\text{U,Pu})\text{O}_2$ [26] and $(\text{U,Er})\text{O}_2$ [27] have been evaluated over a wide temperature, above room temperature, the calculated results are almost in agreement with the experimental values. Arima et al. [20] carried out MD simulations of hypostoichiometric tertiary MOX-fuel $\text{U}_{0.8}\text{Pu}_{0.2}\text{O}_{2-x}$ ($x = 0.0 - 0.06$) solid solutions in the temperature range from 300 to 2000K. Their MD simulation results indicate that the oxygen deficiency x has a larger effect on the thermal conductivity. However, their simulation just in a small temperature (300 – 2000K) and oxygen deficiency (0.0–0.06) range, there is currently no reliable resolution about the effect of oxygen vacancy on the physical properties of MOX. Many publications have reported the thermal properties of actinide dioxides systematically using the molecular dynamics calculations with respective interatomic potential models, including UO_2 [28].

In this work, additional calculations and MD simulations were employed to study the quaternary (U, Pu, Np) O_2 system. This is very significant in this work towards

understanding the effect of a non-uniform lattice in the degradation of the thermal conductivity of mixed oxides and with high burnup. Ideally the analytical equations used to describe defect-phonon scattering in the ternary systems are transferable to the specified quaternary systems, so that predictions can begin to be made for any combination of actinide oxides. Emphasis is placed on comparing through the comparison with excessive plutonium content material; it is proposed that the impact of the O/M should be taken under consideration when predicting the thermal conductivity of combined oxide fuels.

COMPUTATIONAL METHOD

MD Simulations are performed by the Large-scale Atomic/Molecular Massively Parallel Simulator (LAMMPS) code from Sandia National Laboratories [29]. Using a random collision model, based on the Ziegler–Biersack–Littmark potential the recoil energy distribution of fission gas atoms is obtained. The potential of Ziegler–Biersack–Littmark (ZBL) [30] is used to simulate the atomic interaction in the (U, Pu, Np) O₂. During the simulation for the MOX, a quantity of uranium cations were replaced with the neptunium cations randomly in the initial structure, keeping the total number of cations and their valence invariable to charge neutrality.

Periodic boundary is applied in all directions. The [1 0 0], [0 1 0] and [0 0 1] are chosen with the application of the uniaxial deformation for the x-axis, the y-axis and z-axis respectively. In order to eliminate the stress in the samples, they are subjected to NPT ensemble (i.e., at fixed number of atoms N, fixed temperature T, and fixed pressure P) to relax 10 ps at the temperature of T. After the finish of the NPT relaxation, the samples are relaxed again by 5 ps under the NVE ensemble (i.e., at fixed number of atoms N, fixed volume V, and fixed energy E) to achieve a thermodynamic equilibrium. Two loading paths (1D along x-axis and 3D) are then employed and the process of compression also uses the NVE ensemble. The interactions are splined to the universal ZBL potential for short ranges. High energy scattering kinematics is determined by the ZBL parts of the interaction potential.

The simulations allowed temperature effects to be included in the fitting procedure, with the variation in lattice parameter as a function of T being considered, starting at 300 K. A Supercell of $8 \times 8 \times 8$ fluorite unit with the MD simulations consisting of 24000 atoms was created.

MD simulations are also carried out on a range with insertion of atomic of the (U, Pu, Np) O₂ solid compositions to investigate the degradation in thermal conductivity due to a non-uniform cation sublattice and the non-equilibrium method (NEQM), as implemented in the LAMMPS.

2.1 Calculation details

Some structures were created by randomly stipulating U, Pu and Np cations in appropriate proportions across a 4a Wyckoff sites of a given fluorite supercell at constant oxygen to metal ratio which represents a stoichiometric composition in the preferred

Quaternary system. Using the non-equilibrium method, as implemented in the Large-scale Atomic/Molecular Massively Parallel Simulator (LAMMPS), atoms transfer kinetic energy in one region of the supercell to another during motion. The 3-D supercell, $8 \times 8 \times 8$ fluorite unit cells, can be split into slabs along the $\hat{z}$ direction. Kinetic Energy Exchange gradually creates a temperature gradient between boundary and the middle of the supercell as shown in Fig. 1.

The factors to consider as a contribution to the phonon mean free path for the (U, Pu, Np) O₂ are like boundary conditions, phonon-phonon scattering and defect-phonon scattering. The change in thermal conductivity due to the phonon-phonon scattering can generally be described by:

$$k = 1(a + bT) \quad (1)$$

where a and b are constants that are fitted to experimental or, in this case, MD data; which is the same fitting function as recommended by the International Atomic Energy Agency (IAEA) [31] for experimental data except with the omission of the small polaron ambipolar contribution that is not captured by the classical empirical potential approach used here) which is directly fitted to the variation of thermal conductivity due to phonon-phonon scattering as a result of temperature influence.

With the flow of heat from the hot region to the cold region, heat flux, J_z , is activated. Defects such as isolated vacancies and interstitials is obtainable when fastest atom in the slab nearest the periodic boundary condition with that of the slowest atom in the central slab concurrently transfers heat across the supercell. A steady state is attained where J_z is equal and opposite to the kinetic energy exchange at equal temperature. Equation (1), calculated from Fourier's law, is used to determine the thermal conductivity, k , of the supercell from J_z and the temperature gradient,

$$2(T_2 - T_1)/L_z \quad (2)$$

$$k = \frac{J_z L_z}{2(T_2 - T_1)} \quad (3)$$

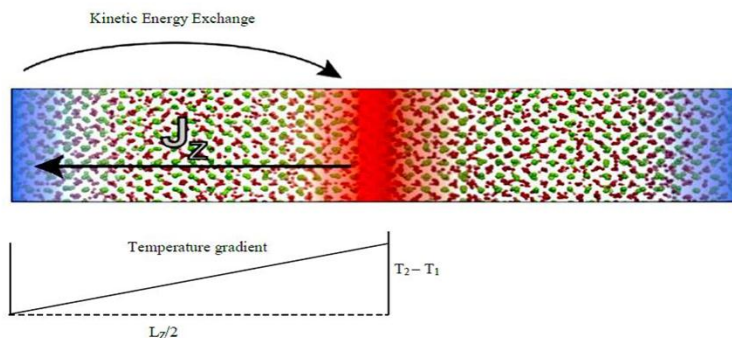


Figure 1: Thermal conductivity during MD uranium and oxygen atoms represents the green and red spheres respectively.

The finite size of the supercell limits the effective phonon mean free path causing an underestimate of thermal conductivity compared to the bulk value. Nonetheless, the thermal conductivity of a finite system, as calculated using the non-equilibrium method, can be examined as a function of the supercell size to determine the bulk value. Derivations from Matthiessen's rule for relaxation time [32], the effective phonon mean free path, l_{eff} , in a finite system can be express as:

$$\frac{1}{l_{eff}} = \frac{1}{l_{\infty}} + \frac{1}{l_{BC}} \quad (4)$$

where l_{∞} is the mean free path in an infinite medium due to both phonon-phonon scattering and, if they are present, scattering from defects. l_{BC} is the mean free path associated with phonon scattering due to boundary conditions. The l_{BC} also represents the average distance travelled by a phonon in the $\hat{z}$ direction between scattering events. The midpoint distance between the hot region and the cold region of the system is written as;

$$\frac{1}{l_{BC}} = \frac{4}{L_z} \quad (5)$$

L_z represents the supercell length in the direction of heat propagation. The thermal conductivity from kinetic theory is [32]:

$$k = \frac{1}{3} v c l_{eff} \quad (6)$$

with c as the specific heat due to the phonons and v is the group velocity of an acoustic branch, the specific heat above the Debye temperature is given by the Dulong-Petit value, such that $c = \frac{3}{2} k_B n$, so that Equations (2)-(4) gives:

$$\frac{1}{k} = \frac{2a^3}{k_B N_c v} \left(\frac{1}{l_{\infty}} + \frac{4}{L_z} \right) \quad (7)$$

where N_c is the number of atoms in the unit cell, a is the lattice parameter and k_B is the Boltzmann constant. An interesting point is that the group velocity of the phonon branch can be approximated as $v = (v_L + 2v_T)/3$, where v_L and v_T are the longitudinal and transverse group velocities respectively.

They are given by $v_L = \sqrt{C_{11}/\rho}$ and $v_T = \sqrt{C_{44}/\rho}$ [32]. For pairwise interactions $C_{11} = C_{44}$, so they are not able to reproduce the correct MOX phonon behavior without underestimating the bulk modulus, $K = \frac{1}{2}(C_{11} + 2C_{12})$. This provides further credence to the choice of the many-body approach used here.

In equation (5) that the inverse of conductivity is a linear function of the inverse of the supercell length. As $L_z \rightarrow \infty$ this yields extrapolation which makes Equation (5)

$$\frac{1}{k} = \frac{2a^3}{k_B N_c v} \left(\frac{1}{l_{\infty}} \right) \quad (8)$$

so that the only contribution to conductivity is from the mean free path associated with phonon-phonon scattering and defect phonon scattering.

RESULTS AND DISCUSSION

The simulation results revealed several important aspects of the (U, Pu, Np) O₂ mechanism. The thermal conductivity (U, Pu, Np) O₂ mixed oxides in the ranges was found to be highly non linear as a function of composition. This is due to the non-uniform cation sublattice as a source of lattice disorder that degrades the thermal conductivity.

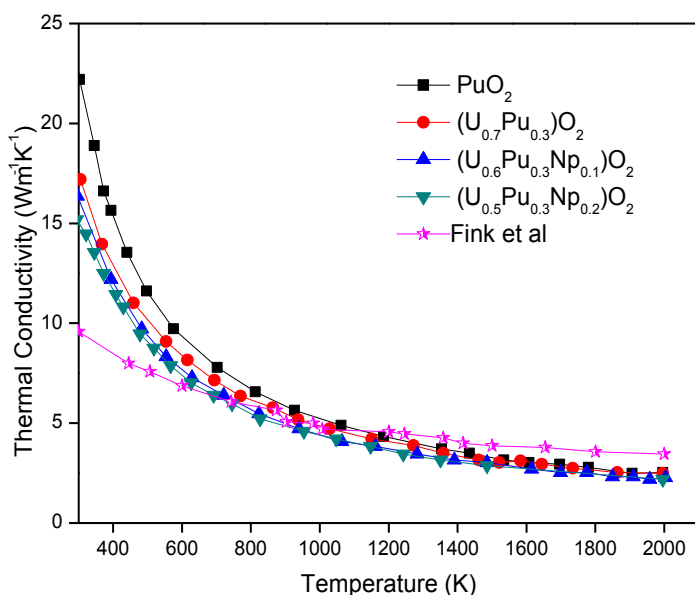


Figure 2: The effects of the addition of neptunium to PuO₂ up to the (U, Pu, Np) O₂ composition as a function of the temperature along with the experimentally obtained value of thermal conductivity [33].

The results for the bulk thermal conductivity for PuO₂, and UO₂, as represented in Figure 2 and Figure 3 respectively is a function of temperature. The thermal conductivity was observed to have reduced slightly because of the reduction in phonon mean free path from scattering which is ideally associated with a non-uniform cation sublattice. Furthermore, Figure 3 shows that the degradation of the UO₂ thermal conductivity due to Pu substitutional defects is relatively small compared to the addition of Np defects. As the higher conductivity of pure PuO₂ begins to exceed the relatively small effect of phonon-defect scattering, the thermal conductivity of (U_{0.5}Pu_{0.3}Np_{0.2})O₂ is higher than that of (U_{0.65}Pu_{0.3}Np_{0.05})O₂ at higher temperatures.

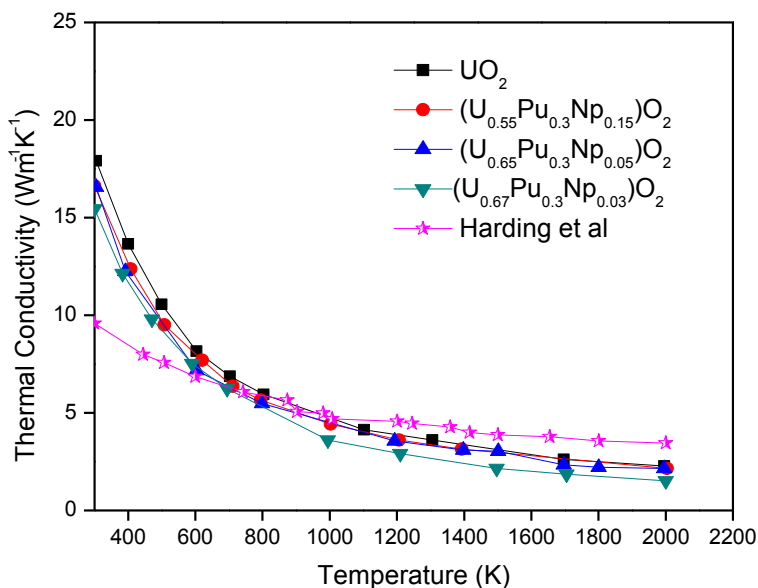


Figure 3: The effect of the addition of neptunium to UO₂ up to the (U, Pu, Np) O₂ composition as a function of the temperature along with the experimentally obtained value of thermal conductivity [34].

Thermal conductivity at various temperatures occur as function of composition. These results were obtained and show only an insignificant reduction in thermal conductivity in the MOX system compared to the limit section. Consequently, this is because of the limited effect of defect scattering in the MOX so that the lower conductivity in pure UO₂ compared to PuO₂ begins to predominate. Also from our work, the thermal conductivities values of the MOX investigated in the different temperature range from 500 K to 3000 K. are shown in Figure 4 can be compared to Morimoto's [25] and Kurosaki's data [35], which is evident that the data plotted in the graph obviously with the increasing temperature, suggests the decrease of thermal conductivities.

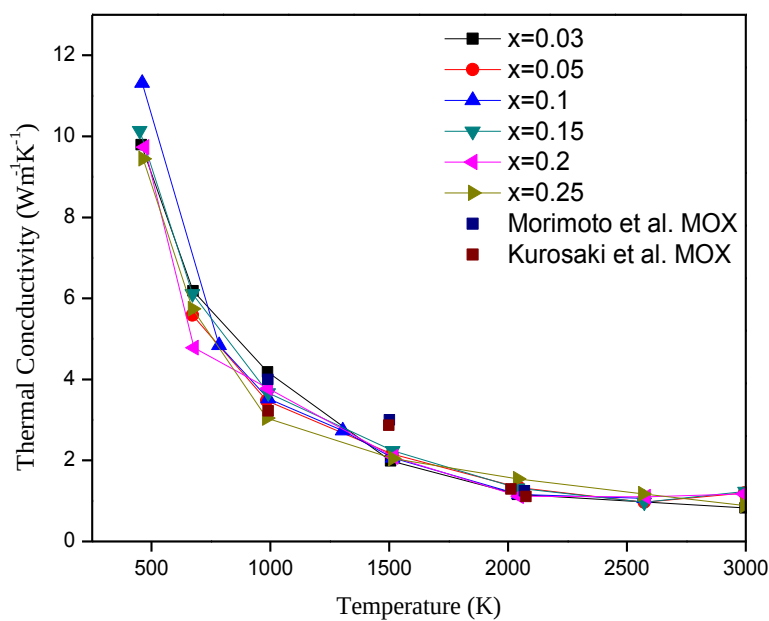


Figure 4: Thermal conductivities calculations by MD simulations vary with different temperatures

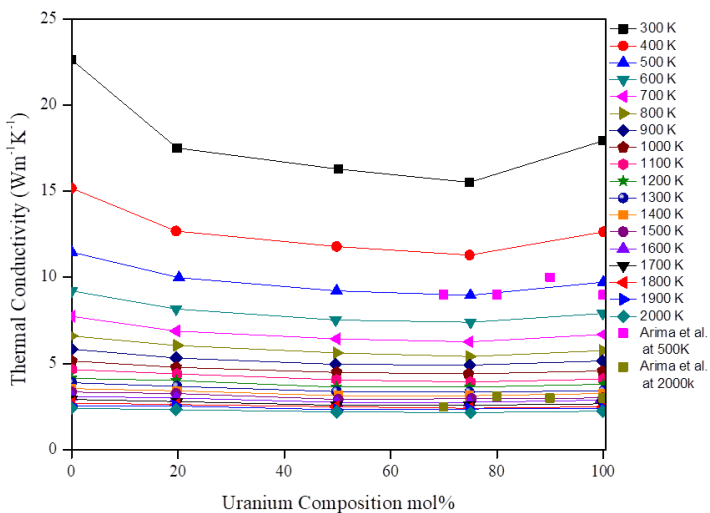


Figure 5: Comparison of the theoretical lattice thermal conductivity of MOX after burnup to Arima et al.

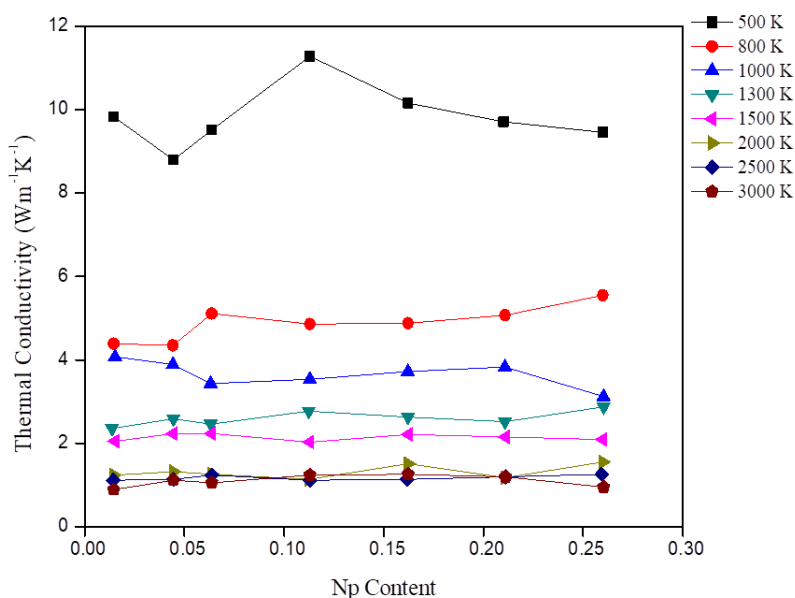


Figure 6: Thermal conductivities calculated by MD simulations vary with different Np content

In Figure 5 the MD results of Arima et al. [20] are also included. This is due to the limited burnup effect of defect scattering in (U, Pu, Np) O₂, and thus, the lower conductivity in pure PuO₂ compared to UO₂ as discussed in the Figure 2 and 3. Figure 6 shows the values of thermal conductivity almost unchanged with several sets of Np concentration. The primary reason is that we increase neptunium ions through substituting uranium ions. The results on the thermal conductivity which is more significant in the (U, Pu, Np) O₂ and also exhibit characteristics that slightly deviates from a linear interpolation of the end member thermal conductivities. In this case the non-uniform cation sublattice contributes to phonon scattering due to substitutional defects. Uranium and plutonium have more similar masses, which then minimises the degradation of the thermal conductivity in (U, Pu, Np) O₂. Similar to oxygen mobility, the degradation of thermal conductivity can be explained in terms of the difference between the end member lattice parameters. This is significant in the MOX and results in stronger phonon scattering by the non-uniform cation sublattice implies more work on cracks, stacks and crystallographic, should be done in the specified MOX Quaternary fuel in order to investigate its physical properties completely.

CONCLUSION

Here we investigate, using atomistic simulation, the thermal conductivity for (U, Pu, Np) O₂ between 300 and 3000 K. The potential ZBL was efficiently employed in the simulation of the atomic interaction in the (U, Pu, Np) O₂. The calculated values of thermal conductivities were also consistent with the literature data which is dependent on the phonon group velocity. The concentration and scattering strength of defects governs the extent to which the phonon mean free path is reduced. With the temperature increasing, the thermal conductivity decreases obviously. Furthermore, the thermal conductivity is hardly affected with the rising neptunium which makes the MOX have good compatibility content. It can be explained from the similar mass and radius between uranium and neptunium. What's more, the supercell size makes little difference compared to the temperature fluctuation. These data will be able to predict the rate and extent of degradation which would be useful to aid in optimising core performance and provide a more complete reference of thermal conductivity of Np containing MOX fuels by MD simulation.

ACKNOWLEDGMENTS

The authors would like to express their sincere thanks to the Harbin Engineering University for innovation-oriented talent research special fund and Heilongjiang academy of sciences who have extended their support.

References

- [1] Ronchi, C., and J. P. Hiernaut. 2004. "Helium Diffusion in Uranium and Plutonium Oxides." *Journal of Nuclear Materials*. <https://doi.org/10.1016/j.jnucmat.2003.10.006>.
- [2] Lu, Yong, Yu Yang, Fawei Zheng, Bao Tian Wang, and Ping Zhang. 2013. "Electronic, Mechanical, and Thermodynamic Properties of Americium Dioxide." *Journal of Nuclear Materials*. <https://doi.org/10.1016/j.jnucmat.2013.06.043>.
- [3] Nakajima, Kuniyoshi, Hiroyuki Serizawa, Noriko Shirasu, Yoshinori Haga, and Yasuo Arai. 2011. "The Solubility and Diffusion Coefficient of Helium in Uranium Dioxide." *Journal of Nuclear Materials*. <https://doi.org/10.1016/j.jnucmat.2011.08.045>.
- [4] Huang, Li, Yilin Wang, and Xi Dai. 2012. "Pressure-Driven Orbital Selective Insulator-to-Metal Transition and Spin-State Crossover in Cubic CoO." *Physical Review B - Condensed Matter and Materials Physics*. <https://doi.org/10.1103/PhysRevB.85.245110>.
- [5] Pipon, Y., C. Raepsaet, D. Roudil, and H. Khodja. 2009. "The Use of NRA to Study Thermal Diffusion of Helium in (U, Pu)O₂." *Nuclear Instruments and Methods in Physics Research, Section B: Beam Interactions with Materials and Atoms*. <https://doi.org/10.1016/j.nimb.2009.03.025>.
- [6] Veshchunov, M. S., R. Dubourg, V. D. Ozrin, V. E. Shestak, and V. I. Tarasov. 2007. "Mechanistic Modelling of Urania Fuel Evolution and Fission Product Migration during 320 Irradiation and Heating." *Journal of Nuclear Materials*. 321 <https://doi.org/10.1016/j.jnucmat.2007.01.081>.

- [7] Gauld, I. C., and U. Mertzyurek. 2019. "Validation of BWR Spent Nuclear Fuel Isotopic Predictions with Applications to Burnup Credit." *Nuclear Engineering and Design*. <https://doi.org/10.1016/j.nucengdes.2019.01.026>.
- [8] Basak, C. B., A. K. Sengupta, and H. S. Kamath. 2003. "Classical Molecular Dynamics Simulation of UO_2 to Predict Thermophysical Properties." *Journal of Alloys and Compounds*. [https://doi.org/10.1016/S0925-8388\(03\)00350-5](https://doi.org/10.1016/S0925-8388(03)00350-5).
- [9] Jackson, R. A., and C. R.A. Catlow. 1985. "Trapping and Solution of Fission Xe in UO_2 . Part 1. Single Gas Atoms and Solution from Underpressurized Bubbles." *Journal of Nuclear Materials*. [https://doi.org/10.1016/0022-3115\(85\)90351-4](https://doi.org/10.1016/0022-3115(85)90351-4).
- [10] Morelon, N. D., D. Ghaleb, J. M. Delaye, and L. Van Brutzel. 2003. "A New Empirical Potential for Simulating the Formation of Defects and Their Mobility in Uranium Dioxide." *Philosophical Magazine*. <https://doi.org/10.1080/1478643031000091454>.
- [11] Geng, Hua Y., Hong X. Song, K. Jin, S. K. Xiang, and Q. Wu. 2011. "First-Principles Study on Oxidation Effects in Uranium Oxides and High-Pressure High-Temperature Behavior of Point Defects in Uranium Dioxide." *Physical Review B - Condensed Matter and Materials Physics*. <https://doi.org/10.1103/PhysRevB.84.174115>.
- [12] Nicoll, S., Hj Matzke, and C. R.A. Catlow. 1995. "A Computational Study of the Effect of Xe Concentration on the Behaviour of Single Xe Atoms in UO_2 ." *Journal of Nuclear Materials*. [https://doi.org/10.1016/0022-3115\(95\)00131-X](https://doi.org/10.1016/0022-3115(95)00131-X).
- [13] Govers, K., S. Lemehov, M. Hou, and M. Verwerft. 2009. "Molecular Dynamics Simulation of Helium and Oxygen Diffusion in UO_{2+x} ." *Journal of Nuclear Materials*. <https://doi.org/10.1016/j.jnucmat.2009.10.043>.
- [14] Miekeley, W., and F. W. Felix. 1972. "Effect of Stoichiometry on Diffusion of Xenon in UO_2 ." *Journal of Nuclear Materials*. [https://doi.org/10.1016/0022-3115\(72\)90080-3](https://doi.org/10.1016/0022-3115(72)90080-3).
- [15] Nogita, K., and K. Une. 1995. "Irradiation-Induced Recrystallization in High Burnup UO_2 Fuel." *Journal of Nuclear Materials*. [https://doi.org/10.1016/0022-3115\(95\)00123-9](https://doi.org/10.1016/0022-3115(95)00123-9).
- [16] Rondinella, Vincenzo V., and Thierry Wiss. 2010. "The High Burn-up Structure in Nuclear Fuel." *Materials Today*. [https://doi.org/10.1016/S1369-7021\(10\)70221-2](https://doi.org/10.1016/S1369-7021(10)70221-2).
- [17] Yamada, Kazuhiro, Ken Kurosaki, Msayoshi Uno, and Shinsuke Yamanaka. 2000. "Evaluation of Thermal Properties of Uranium Dioxide by Molecular Dynamics." *Journal of Alloys and Compounds*. [https://doi.org/10.1016/S0925-8388\(00\)00806-9](https://doi.org/10.1016/S0925-8388(00)00806-9).
- [18] Arima, Tatsumi, Sho Yamasaki, Yaohiro Inagaki, and Kazuya Idemitsu. 2005. "Evaluation of Thermal Properties of UO_2 and PuO_2 by Equilibrium Molecular Dynamics Simulations from 300 to 2000 K." *Journal of Alloys and Compounds*. <https://doi.org/10.1016/j.jallcom.2005.04.003>.
- [19] Arima, Tatsumi, Sho Yamasaki, Yaohiro Inagaki, and Kazuya Idemitsu. 2006. "Evaluation of Thermal Conductivity of Hypostoichiometric (U, Pu) O_{2-x} Solid Solution by Molecular Dynamics Simulation at Temperatures up to 2000 K." *Journal of Alloys and Compounds*. <https://doi.org/10.1016/j.jallcom.2005.08.003>.
- [20] Nichenko, S., and D. Staicu. 2014. "Thermal Conductivity of Porous UO_2 : Molecular Dynamics Study." *Journal of Nuclear Materials*. 299 <https://doi.org/10.1016/j.jnucmat.2014.08.009>.
- [21] Tijo Vazhappilly, Arup Kumar Pathak. 2019. "Theoretical study on the mechanical and thermal properties of uranium dioxide doped with lanthanide fission products." *Journal of Nuclear Materials*. 519 <https://doi.org/10.1016/j.jnucmat.2019.03.032>.
- [22] Mingjie, Wan, Zhang Li, Du Jiguang, Huang Duohui, Wang Lili, and Jiang Gang. 2012. "The MD Simulation of Thermal Properties of Plutonium Dioxide." *Physica B: Condensed Matter*. <https://doi.org/10.1016/j.physb.2012.08.010>.
- [23] Juan J Carbajo, Gradyon L Yoder, Sergey G Popov, Victor K Ivanov. 2001. "A review of the thermophysical properties of MOX and UO_2 fuels." *Journal of Nuclear Materials*, 299. [https://doi.org/10.1016/S0022-3115\(01\)00692-4](https://doi.org/10.1016/S0022-3115(01)00692-4).
- [24] Ma, Li, and Asok K. Ray. 2012. "Formation Energies and Swelling of Uranium Dioxide by Point Defects." *Physics Letters, Section A: General, Atomic and Solid State Physics*. <https://doi.org/10.1016/j.physleta.2012.03.017>.

- [25] Morimoto, Kyoichi, Masato Kato, Masahiro Ogasawara, Motoaki Kashimura, and Tomoyuki Abe. 2008. "Thermal Conductivities of (U, Pu, Am)O₂ Solid Solutions." *Journal of Alloys and Compounds*. <https://doi.org/10.1016/j.jallcom.2006.12.159>.
- [26] Schelling, Patrick K., Simon R. Phillpot, and Pawel Keblinski. 2002. "Comparison of Atomic-Level Simulation Methods for Computing Thermal Conductivity." *Physical Review B - Condensed Matter and Materials Physics*. <https://doi.org/10.1103/PhysRevB.65.144306>.
- [27] Shinsuke Yamanaka, Ken Kurosaki, Masahito Katayama, Jun Adachi, Masayoshi Uno, Takeshi Kuroishi, Masatoshi Yamasaki. 2009. Thermal and mechanical properties of (U,Er)O₂." *Journal of Nuclear Materials*. 389. <https://doi.org/10.1016/j.jnucmat.2009.01.016>
- [28] Liu X.-Y., M. W. D. Cooper, K. J. McClellan, J. C. Lashley, D. D. Byler, B. D. C. Bell, R. W. Grimes, C. R. Stanek, and D. A. Andersson. 2016. "Molecular Dynamics Simulation of Thermal Transport in UO₂ Containing Uranium, Oxygen, and Fission-product Defects." *Phys. Rev. Applied* 6, 044015
- [29] Plimpton, S. 1995. "Fast Parallel Algorithms for Short – Range Molecular Dynamics." *Journal of Computational Physics* 117 (June 1994): 1–19. <https://doi.org/10.1006/jcph.1995.1039>.
- [30] Biersack, J.P., and J.F. Ziegler. 1985. "The Stopping and Range of Ions in Solids." *Ion Implantation Techniques Springer Series in Electrophysics Volume 10*, no. Volume 1: 122–56. https://doi.org/10.1007/978-3-642-68779-2_5.
- [31] IAEA. 2006. "Thermophysical Properties Database of Materials for Light Water Reactors and Heavy Water Reactors." *IAEA-TECDOC*.
- [32] Patrick K. Schelling, Simon R. Phillpot, P. Keblinski, "Comparison of atomic-level simulation methods for computing thermal conductivity." *Phys. Rev. B* 65 (2002) 144306. <https://doi.org/10.1103/PhysRevB.65.144306>
- [33] Fink J.K., 2000. "Thermophysical properties of uranium dioxide." *Journal of Nuclear Materials*. 279; 1-18.
- [34] Harding J.H., Martin D.G. 1989. "A recommendation for the thermal conductivity of UO₂." *Journal of Nuclear Materials*. 166; 223-226. [https://doi.org/10.1016/0022-3115\(89\)90218-3](https://doi.org/10.1016/0022-3115(89)90218-3)
- [35] Kurosaki, Ken, Jun Adachi, Masahito Katayama, Masahiko Osaka, Kenya Tanaka, Masayoshi Uno, and Shinsuke Yamanaka. 2006. "Molecular Dynamics Studies of Americium-275 Containing Mixed Oxide Fuels." *Journal of Nuclear Science and Technology*. <https://doi.org/10.1080/18811248.2006.9711215>.